

## INDIGA, Russian Federation

WMO: 222920

Lat: 67.689N

Lon: 48.684E

Elev: 14

StdP: 14.69

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | -26.0      | -21.5 | -31.4                         | 0.9 | -25.8 | -26.7 | 1.2 | -21.4 | 46.4                  | 12.8 | 41.3 | 18.5 | 6.1                      | 110  | 0.970 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.0                   | 70.9            | 61.8 | 66.9 | 59.7 | 63.5 | 58.2 | 63.2                | 69.2 | 61.1 | 65.8 | 58.6 | 62.5 | 12.5                 | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 60.8                            | 79.6 | 66.5 | 58.7 | 73.7 | 63.9 | 56.3 | 67.5 | 61.7 | 28.7          | 69.0 | 27.2 | 66.1 | 25.5 | 62.5 | 70.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 35.9              | 31.9 | 28.7 | -30.6                      | 77.4 | 4.2                | 3.7 | -33.6                                              | 80.1 | -36.1      | 82.2 | -38.4      | 84.3 | -41.5      | 87.1 |    |    |
|                   |      |      | -30.8                      | 66.0 | 4.2                | 3.0 | -33.7                                              | 68.1 | -36.2      | 69.9 | -38.5      | 71.6 | -41.5      | 73.7 |    |    |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 29.4   | 7.3   | 11.4  | 14.7  | 24.5  | 33.0  | 43.9  | 52.8  | 51.3  | 44.7  | 34.0  | 20.6  | 13.9  |
|                                                                   | DBStd       | 18.64  | 14.29 | 14.83 | 12.79 | 8.69  | 8.20  | 7.83  | 8.15  | 5.80  | 5.53  | 7.14  | 11.69 | 13.80 |
|                                                                   | HDD50       | 7798   | 1323  | 1080  | 1094  | 766   | 533   | 219   | 64    | 50    | 174   | 495   | 882   | 1118  |
|                                                                   | HDD65       | 12988  | 1788  | 1500  | 1559  | 1216  | 994   | 634   | 386   | 426   | 610   | 960   | 1332  | 1583  |
|                                                                   | CDD50       | 296    | 0     | 0     | 0     | 0     | 4     | 35    | 152   | 91    | 14    | 0     | 0     | 0     |
|                                                                   | CDD65       | 10     | 0     | 0     | 0     | 0     | 0     | 1     | 8     | 1     | 0     | 0     | 0     | 0     |
|                                                                   | CDH74       | 42     | 0     | 0     | 0     | 0     | 0     | 3     | 35    | 4     | 0     | 0     | 0     | 0     |
|                                                                   | CDH80       | 1      | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0     |
|                                                                   | WSAvg       | 14.3   | 15.6  | 15.6  | 15.2  | 14.3  | 12.8  | 12.5  | 11.9  | 12.9  | 13.5  | 15.7  | 16.1  | 16.0  |
| Precipitation                                                     | PrecAvg     | 16.4   | 1.3   | 1.0   | 0.8   | 0.7   | 0.8   | 1.7   | 1.7   | 2.0   | 2.0   | 2.2   | 1.1   | 0.9   |
|                                                                   | PrecMax     | 27.8   | 4.0   | 3.0   | 1.7   | 1.4   | 3.0   | 5.2   | 5.3   | 4.6   | 4.3   | 5.1   | 1.9   | 2.3   |
|                                                                   | PrecMin     | 8.6    | 0.3   | 0.1   | 0.1   | 0.1   | 0.2   | 0.2   | 0.1   | 0.4   | 0.7   | 0.7   | 0.6   | 0.3   |
|                                                                   | PrecStd     | 4.0    | 0.9   | 0.6   | 0.4   | 0.4   | 0.5   | 1.1   | 1.1   | 1.1   | 1.0   | 1.0   | 0.4   | 0.5   |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 32.3  | 34.1  | 34.8  | 44.8  | 63.8  | 72.0  | 78.4  | 72.7  | 61.7  | 49.1  | 37.0  | 34.5  |
|                                                                   |             | MCWB   | 31.4  | 33.4  | 33.1  | 39.8  | 53.5  | 61.3  | 64.4  | 64.2  | 56.8  | 46.9  | 35.6  | 33.3  |
|                                                                   | 2%          | DB     | 30.5  | 32.6  | 33.5  | 38.5  | 55.9  | 66.0  | 73.2  | 67.4  | 57.9  | 46.3  | 35.8  | 33.2  |
|                                                                   |             | MCWB   | 29.5  | 32.1  | 32.4  | 35.4  | 49.5  | 57.3  | 63.1  | 60.8  | 54.0  | 44.9  | 34.7  | 32.4  |
|                                                                   | 5%          | DB     | 28.1  | 30.8  | 32.0  | 36.1  | 49.4  | 61.2  | 69.0  | 64.0  | 55.0  | 44.4  | 34.4  | 32.0  |
|                                                                   |             | MCWB   | 27.3  | 30.3  | 31.1  | 34.0  | 44.6  | 54.7  | 61.3  | 59.2  | 52.0  | 42.9  | 33.4  | 31.3  |
|                                                                   | 10%         | DB     | 26.0  | 28.9  | 30.1  | 34.5  | 44.1  | 56.5  | 65.5  | 60.6  | 52.3  | 42.6  | 33.3  | 30.2  |
|                                                                   |             | MCWB   | 25.2  | 28.1  | 29.3  | 33.1  | 40.8  | 51.9  | 59.6  | 57.3  | 50.2  | 41.0  | 32.4  | 29.6  |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 31.8  | 33.6  | 33.6  | 40.3  | 54.5  | 62.8  | 66.5  | 65.8  | 58.1  | 47.6  | 36.3  | 33.9  |
|                                                                   |             | MCDB   | 32.1  | 33.8  | 34.1  | 44.8  | 61.8  | 69.5  | 73.8  | 72.1  | 60.9  | 48.4  | 36.7  | 34.0  |
|                                                                   | 2%          | WB     | 29.7  | 32.2  | 32.6  | 35.6  | 49.4  | 58.0  | 64.2  | 62.0  | 54.9  | 45.0  | 34.8  | 32.7  |
|                                                                   |             | MCDB   | 30.4  | 32.5  | 33.3  | 37.4  | 56.0  | 63.4  | 71.3  | 66.1  | 57.1  | 46.0  | 35.3  | 33.1  |
|                                                                   | 5%          | WB     | 27.6  | 30.3  | 31.2  | 34.3  | 44.7  | 54.9  | 62.4  | 59.8  | 52.2  | 43.1  | 33.6  | 31.3  |
|                                                                   |             | MCDB   | 28.3  | 30.7  | 31.9  | 35.7  | 48.9  | 60.6  | 68.2  | 63.2  | 54.2  | 44.2  | 34.1  | 31.9  |
|                                                                   | 10%         | WB     | 25.3  | 28.2  | 29.4  | 33.2  | 40.9  | 51.8  | 60.2  | 57.3  | 50.4  | 41.2  | 32.5  | 29.6  |
|                                                                   |             | MCDB   | 26.1  | 28.9  | 30.1  | 34.4  | 44.1  | 56.8  | 64.6  | 60.4  | 52.2  | 42.4  | 33.2  | 30.2  |
| Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 12.2  | 10.6  | 11.0  | 9.9   | 9.2   | 10.3  | 10.0  | 9.1   | 7.4   | 6.2   | 8.8   | 10.4  |
|                                                                   |             | MCDBR  | 11.2  | 7.4   | 5.9   | 10.0  | 19.4  | 20.1  | 17.7  | 15.8  | 11.9  | 6.7   | 6.8   | 6.6   |
|                                                                   |             | MCWBR  | 11.4  | 7.5   | 5.7   | 8.2   | 12.9  | 11.8  | 8.8   | 10.3  | 8.8   | 6.4   | 6.2   | 6.8   |
|                                                                   | 5% WB       | MCDBR  | 11.3  | 7.9   | 5.8   | 8.9   | 19.3  | 18.4  | 16.1  | 14.2  | 10.4  | 6.4   | 6.5   | 6.5   |
|                                                                   |             | MCWBR  | 11.4  | 7.9   | 5.7   | 7.6   | 13.1  | 12.5  | 8.9   | 10.2  | 8.0   | 6.4   | 6.5   | 6.8   |
| Clear-Sky Solar Irradiance                                        | taub        |        | 0.213 | 0.258 | 0.299 | 0.329 | 0.339 | 0.341 | 0.375 | 0.353 | 0.324 | 0.288 | 0.366 | N/A   |
|                                                                   | taud        |        | 1.903 | 2.030 | 2.087 | 2.147 | 2.331 | 2.477 | 2.413 | 2.459 | 2.491 | 2.248 | 2.581 | N/A   |
|                                                                   | Ebn at Noon |        | 31    | 168   | 229   | 258   | 271   | 274   | 258   | 252   | 231   | 160   | 38    | 0     |
|                                                                   | Edh at Noon |        | 3     | 16    | 28    | 35    | 33    | 29    | 30    | 26    | 20    | 13    | 4     | 0     |
| All-Sky Solar Radiation                                           | RadAvg      |        | 12    | 154   | 620   | 1193  | 1534  | 1611  | 1530  | 1004  | 521   | 163   | 27    | 0     |
|                                                                   | RadStd      |        | 1     | 20    | 62    | 93    | 88    | 194   | 181   | 102   | 58    | 26    | 3     | 0     |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (3 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page